Kedhar Swaminathan

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SUMMARY

Economics graduate and Springboard Data Analytics fellow who builds end-to-end data pipelines in Python and SQL, turning messy source data into analysis that supports a decision. Seeking an entry-level Data Analyst role.

EDUCATION

B.S. Economics — University of Houston May 2025

- Coursework: Introduction to Economic Data Analysis (Excel, RStudio, Jupyter), Econometrics, Principles of Micro- and Macroeconomics.

Data Analytics Career Track — Springboard In Progress, 2026

- Mentor-led program covering SQL, Python, statistics, and machine learning, built around capstone projects on real datasets.

TECHNICAL SKILLS

Languages: Python (pandas, NumPy, scikit-learn), SQL (Microsoft SQL, DuckDB), R

Data Engineering: ETL pipeline development, data cleaning and normalization, categorical standardization, data validation, reproducible analysis workflows

Analysis: classification modeling, model evaluation (ROC-AUC, precision/recall), cost-benefit and sensitivity analysis, descriptive statistics, hypothesis testing

Tools: Git/GitHub, Jupyter Notebook, VS Code, Excel (pivot tables, lookups), matplotlib/seaborn

TECHNICAL PROJECTS

DFW Hospital Pricing Transparency Analysis | Python, DuckDB, SQL 2026

- Built a Python/DuckDB pipeline ingesting federally mandated price transparency files from five Dallas-Fort Worth hospitals to test whether published pricing enables meaningful patient cost comparison.
- Built a rule-based payer classification cascade — plan-name regex, payer-group defaults, payer-name fallback — refined across four versions to catch failures such as Medicare PFFS plans misfiling as traditional Medicare, cutting unclassified payer/plan combinations from 46% to 1.8%.
- Found the same outpatient knee MRI varies 2.1x in negotiated commercial rates across four DFW hospitals (3.7x to 8.0x Medicare), and traced a fifth hospital's apparent 16x outlier to percent-of-charges contracts — evidence published files cannot support like-for-like comparison.

Customer Churn Prediction & Retention Cost Model | Python, scikit-learn 2026

- Built a classification pipeline on 7,043 telecom customer records (21 variables), handling type coercion errors, categorical encoding, and feature standardization with leakage-safe train/test separation.
- Compared logistic regression, random forest, and gradient boosting; selected logistic regression (0.835 ROC-AUC) over the marginally higher-scoring gradient boosting for coefficient interpretability, and raised recall from 57% to 79% through class weighting.
- Modeled the retention decision explicitly rather than accepting the default 0.5 classification threshold, showing the cost-optimal cutoff shifts from 0.69 to 0.14 across plausible customer-lifetime-value assumptions — an eightfold swing in campaign size — and concluded no threshold could be recommended without finance data.

Nintendo Game Reception Prediction | Python, pandas, scikit-learn 2026

- Built a classification pipeline on 706 Nintendo titles predicting critical reception from game characteristics alone — platform, genre, developer, ESRB rating, and release year — comparing three classifiers against a majority-class baseline (best ROC-AUC 0.72 vs. 0.500).
- Diagnosed target leakage in which the outcome variable remained in the feature matrix due to a silently suppressed drop error, and removed 385 records whose labels were fabricated from missing values; rebuilt the pipeline from 3,421 columns to 51 and re-reported results honestly.

EXPERIENCE

Fundraising & Data Intern (Part-Time) — Indian Institute of Science Jul 2025 – Dec 2025

- Managed fundraising accounts and performed queries and data analysis to maintain detailed records of donors and contribution amounts, generating insights to support decision-making.
- Researched and analyzed collected data to support ongoing projects and fundraising for a research hospital planned for the campus.